Powers with a rational exponent

One definition joins roots and powers into a single system — and all the old index laws keep working.

LESSON 30–31 | 2 × 40 MIN | ALGEBRA 8, §9 | STAGE 9 · 1.3

LESSON CLOCK

5' | 12' | 8' | 13' | 2'

Warm-up	5 min
Explanation	12 min
Interactive	8 min
Practice	13 min
Homework	2 min

LEARNING OBJECTIVES

- Define $a^{1/n}$ and $a^{m/n}$.
- Convert freely between root notation and index notation.
- Use the index laws with fractional and negative exponents.
- Evaluate powers with a rational exponent without a calculator.

TEXTBOOK

National	§9, pp. 42–48
Cambridge	Learner's Book pp. 16–20
Workbook	Workbook 1.3

01 Explanation

The definition

DEFINITION

For $a > 0$ and natural m, n with $n \geq 2$:

$$a^{1/n} = \sqrt[n]{a} \quad \text{and} \quad a^{m/n} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$$

and, as before, $a^0 = 1$ and $a^{-k} = \frac{1}{a^k}$.

Why this definition and no other? Because it is the only one that keeps the index laws true. If $(a^{1/2})^2 = a^1 = a$ then $a^{1/2}$ must be the number that squares to a — that is $\sqrt{a}$.

The **denominator of the index is the root**, the **numerator is the power**. Say it out loud once and the whole topic becomes bookkeeping.

The index laws, unchanged

$$a^p \cdot a^q = a^{p+q} \quad \cdot \quad a^p : a^q = a^{p-q} \quad \cdot \quad (a^p)^q = a^{pq}$$

$$(ab)^p = a^p b^p \quad \cdot \quad \left(\frac{a}{b}\right)^p = \frac{a^p}{b^p}$$

The letters p and q are now any rational numbers, positive or negative. Nothing else changes.

Evaluating without a calculator

Take the **root first**, then the power — the numbers stay small.

$$8^{2/3} = (\sqrt[3]{8})^2 = 2^2 = 4$$

Going the other way, $\sqrt[3]{8^2} = \sqrt[3]{64} = 4$ — same answer, bigger arithmetic.

NEGATIVE INDEX

a^{-p} means **one over** a^p; it does not make the answer negative. $4^{-1/2} = \frac{1}{4^{1/2}} = \frac{1}{2}$.

02 Worked examples

EXAMPLE 1 ***Evaluate*** $27^{2/3}$

1. Denominator 3 → cube root; numerator 2 → square.
   Root first.
2. $\sqrt[3]{27} = 3$
3. $3^2 = 9$

ANSWER

9

EXAMPLE 2 **Evaluate $16^{-3/4}$**

1. $16^{-3/4} = \dfrac{1}{16^{3/4}}$

   A negative index means a reciprocal.

2. $\sqrt[4]{16} = 2$

   Root first.

3. $2^3 = 8$

4. $= \dfrac{1}{8}$

ANSWER

$\dfrac{1}{8}$

EXAMPLE 3 **Simplify $a^{1/2} \cdot a^{1/3}$**

1. $a^{1/2} \cdot a^{1/3} = a^{1/2 + 1/3}$

   Add the indices.

2. $\dfrac{1}{2} + \dfrac{1}{3} = \dfrac{5}{6}$

   Common denominator 6.

3. $= a^{5/6}$

   or $\sqrt[6]{a^5}$

ANSWER

$a^{5/6}$

03 Interactive model

Set n = 3, m = 2, a = 8 and read both notations giving 4.

Root notation and index notation

$$\sqrt[3]{64} = 64^{1/3} = 4$$

base a **64**	index n **3**	power m **1**	check: value^3 **64**	a^1 **64**

Raising the answer to the power 3 gives back a^1 — that is exactly what “root of degree 3” means.

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	OʻZBEKCHA	РУССКИЙ
Rational exponent	Ratsional koʻrsatkich	Рациональный показатель
Power	Daraja	Степень
Base	Asos	Основание
Negative index	Manfiy koʻrsatkich	Отрицательный показатель
Zero index	Nol koʻrsatkich	Нулевой показатель
Index law	Daraja xossasi	Свойство степени
Reciprocal	Teskari son	Обратное число
Equivalent form	Ekvivalent koʻrinish	Равносильная форма
Evaluate	Qiymatini topish	Вычислить

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

$9^{1/2}$ equals:

- 4.5
- 3
- 81
- 18

$8^{2/3}$ equals:

- 4
- 16
- $\frac{16}{3}$
- 6

$a^{-1/2}$ equals:

- $-\sqrt{a}$
- $\frac{1}{\sqrt{a}}$
- $\sqrt{-a}$
- a^2

$a^{1/2} \cdot a^{1/2}$ equals:

- $a^{1/4}$
- a
- $2a^{1/2}$
- a^2

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. Write $\sqrt{a}$ as a power

ANSWER $a^{1/2}$

2. Write $\sqrt[3]{a}$ as a power

ANSWER $a^{1/3}$

3. Write $a^{1/4}$ as a root

ANSWER $\sqrt[4]{a}$

4. Evaluate $9^{1/2}$

ANSWER 3

5. Evaluate $8^{1/3}$

ANSWER 2

6. Evaluate $16^{1/2}$

ANSWER 4

7. Evaluate 5^0

ANSWER 1

Medium · the standard question

7 PROBLEMS

1. *Evaluate* $27^{2/3}$

   ANSWER 9

2. *Evaluate* $16^{3/4}$

   ANSWER 8

3. *Evaluate* $4^{-1/2}$

   ANSWER $\frac{1}{2}$

4. *Simplify* $a^{1/2} \cdot a^{1/3}$

   ANSWER $a^{5/6}$

5. *Simplify* $a^{3/4} : a^{1/4}$

   ANSWER $a^{1/2}$

6. *Simplify* $(a^{2/3})^3$

   ANSWER a^2

7. *Evaluate* $25^{3/2}$

   ANSWER 125

Hard · stretch and reason

7 PROBLEMS

1. *Evaluate* $16^{-3/4}$

   ANSWER $\frac{1}{8}$

2. *Evaluate* $32^{2/5}$

   ANSWER 4

3. *Simplify* $(a^{1/2}b^{1/3})^6$

   ANSWER a^3b^2

4. *Simplify* $\frac{a^{5/6}}{a^{1/3}}$

   ANSWER $a^{1/2}$

5. *Evaluate* $(\frac{1}{8})^{-2/3}$

   ANSWER 4

6. *Write* $\sqrt{a} \cdot \sqrt[3]{a}$ *as one power*

   ANSWER $a^{5/6}$

7. *Evaluate* $81^{0.75}$

   ANSWER $81^{3/4} = 27$

07 Homework

Homework — 6 problems

Algebra 8, §9, pp. 42–48. Take the root first, then the power.

1. *Evaluate* $64^{1/3}$ *and* $64^{2/3}$
2. *Evaluate* $100^{1/2}$ *and* $100^{-1/2}$
3. *Simplify* $a^{2/5} \cdot a^{3/5}$
4. *Simplify* $(b^{3/4})^8$

5. *Evaluate* $125^{2/3}$

6. *Write* $\sqrt[4]{a^3}$ *as a power with a rational index*